

Codebase Explanation: The Function Dictionary

1. The Welcome & Setup Engine

These functions handle the user's first impression and set up the page.

- **saveUsername()**
 - **What it does:** Grabs the name the user typed into the welcome gate. If it's valid, it saves it to the browser's memory (localStorage). It then hides the gate and plays the cinematic "WELCOME [NAME]" text animation before revealing the main website.
- **updateHeader(name)**
 - **What it does:** Simply takes the saved name and displays it in the top left corner of the site as "Viewing as: [Name]".
- **setupLensInteractions()**
 - **What it does:** The "Lens" is your custom amber circle cursor on desktop. This function attaches "listeners" to every button, image, and link on the site. When the mouse hovers over them, it tells the lens to grow larger, change color, and display the word "view".

2. The Visual Gallery Engine

These functions build and manage the main photo feed.

- **buildGalleryGrid()**
 - **What it does:** The heavy lifter. It loops through your massive myImagesCollection array and writes the HTML for every single photo card, inserting the image, title, EXIF data, and the Like/Comment buttons automatically.
- **runScrollObserver()**

- **What it does:** Acts as a lookout. It watches the screen as the user scrolls. When a hidden image card is about to enter the screen, it adds a "visible" class to trigger the smooth slide-up and fade-in animation.
- **toggleGalleryTheme()**
 - **What it does:** Acts as a light switch. It flips a variable between true/false and swaps the CSS colors from the dark "Darkroom" theme to the bright "Daylight" theme.
- **launchFocusedView(src, text) & dismissFocusedView()**
 - **What it does:** Opens and closes the "Lightbox". When a user clicks a photo, it grabs the high-res URL and EXIF data, throws up a black blurred background, and displays the image full-screen.

3. The Audio & Music Player

These functions manage the continuous background soundtrack.

- **assemblePlaylistUI()**
 - **What it does:** Reads your 44-song myPlaylist array and generates the clickable list of tracks inside the Music side-drawer.
- **runAudioSourceTrack(id)**
 - **What it does:** The disc jockey. It receives a song ID, loads the MP3 link from Dropbox into the hidden <audio> tag, updates the "Now Playing" text on the screen, and starts playing the music.
- **toggleAudioPlayback()**
 - **What it does:** Checks if the music is currently playing or paused. If playing, it pauses it; if paused, it resumes. It also toggles the Play/Pause SVG icons in the UI.
- **playNextTrack() & playPrevTrack()**

- **What it does:** Calculates what song is currently playing, adds or subtracts 1 to find the next/previous song, and tells the DJ to play it. If it hits the end of the playlist, it intelligently loops back to the beginning.
- **formatTime(seconds)**
 - **What it does:** A math helper. It takes raw seconds (e.g., 125) and converts it into a human-readable music timestamp (e.g., 2:05) for your progress slider.

4. Supabase Engagement (Likes & Comments)

These functions talk to your cloud database to keep the site updated for everyone.

- **loadGlobalStats()**
 - **What it does:** Runs immediately when the site loads. It asks Supabase, "Give me all the like counts and comment counts for every photo," and then magically updates all the little numbers below the pictures.
- **triggerGlobalLike(event, photoid)**
 - **What it does:** Handles the "Like" button click. First, it checks sessionStorage to ensure the user hasn't liked more than 10 photos this visit. If they are under the limit, it fires the confetti/pigeon animations and tells Supabase to permanently add +1 to that photo's global score.
- **safeFallbackLikeUpdate(photoid)**
 - **What it does:** A safety net. If the primary method of updating likes fails (due to a database hiccup), this function manually fetches the old score, adds 1, and forces an update.
- **triggerDoubleTapLike(element, photoid)**

- **What it does:** Listens for a double-click on a photo. It triggers a quick "Heart Flash" visual over the image, and then silently runs the global like function.
- **loadAndOpenComments(photoid)**
 - **What it does:** Opens the right-side comments drawer. Before showing it, it asks Supabase for all the historical comments left on that specific photo, formats the dates, and displays them in a scrollable feed.
- **submitNewCardComment()**
 - **What it does:** Grabs the text the user typed and their saved name, sends it to Supabase for permanent storage, and instantly injects a green-highlighted version of their comment into the feed so they see it immediately.

5. The Arcade & Mini-Games

These functions power the hidden games menu.

- **switchGameMode(mode)**
 - **What it does:** Hides and shows the different game screens (Tic-Tac-Toe, Sudoku, Snake) depending on which top tab the user clicks.
- **makeTTTMove(index) & executeMove() & checkTTTGameStatus()**
 - **What it does:** The Tic-Tac-Toe manager. It lets the player place an 'X'. Then it checks if the player won. If not, it pauses for a fraction of a second ("Shirsendu is thinking...") and triggers the AI.
- **calculateAICounterMove()**
 - **What it does:** The "Brain" of Tic-Tac-Toe. It checks if the computer can win this turn. If not, it checks if it needs to block the player from

winning. If neither, it picks the center square or a random empty spot.

- **generateNewSudoku9x9() & checkSudoku9x9Solution()**
 - **What it does:** Draws a 9x9 grid, filling in specific blocks with unchangeable numbers, and leaving others as blank text inputs. The check function ensures the user's typed numbers exactly match the hidden master solution array.
- **prepareSnakeBoard(), startSnakeActiveLoop(), & renderSnakeGameTickFrame()**
 - **What it does:** The Snake engine. It draws the starting snake on an HTML Canvas. The "Loop" function fires every 160 milliseconds, erasing the old canvas, moving the snake's coordinates forward based on keyboard arrows, checking if it ate the food (to grow and add points), and checking if it hit a wall or its own tail (Game Over).

6. The Special Tab (Vault)

These functions protect and manage the secret hidden page.

- **validateVaultAccess()**
 - **What it does:** The security guard. Checks if the user typed 05042001. If wrong, it flashes a red error. If right, it handles a 9-second multi-stage cinematic animation: hiding the gate -> showing a welcome message -> showing the heart timer -> sliding in the secret photos.
- **startHeartLiveTimer()**
 - **What it does:** A continuous loop that subtracts December 2, 2024 from the exact current time. It calculates the remaining Days, Hours, Minutes, and Seconds and updates the pulsing heart numbers every 1 second.
- **triggerCardStaggerCascade()**

- **What it does:** A visual flair function. Instead of all secret photos appearing at once, it loops through them and adds a slight delay to each one, making them cascade onto the screen one by one.

7. Special Effects & Cinematic Events

The "Wow Factor" functions that run in the background.

- **triggerConfettiBurst(x, y) & stepConfettiSimulationFrame()**
 - **What it does:** When called, it generates 35 tiny colored squares at the user's mouse coordinates. The simulation frame runs 60 times a second, applying fake "gravity" to make the squares arc upward and slowly drift down while fading away.
- **launchRealisticPigeons(), createPigeon(), & createFeather()**
 - **What it does:** Plays an audio clip of wings flapping. It spawns an image of a pigeon off-screen and uses sine-wave math to make it "fly" across the screen with a realistic bobbing/flapping motion, occasionally dropping a blurred CSS feather.
- **acceptCongrats()**
 - **What it does:** After a user sits on the website for 10 minutes, a pop-up appears. When they click "Acknowledge", this function closes the pop-up, triggers the Pigeon flight, and fires off 5 massive blasts of confetti to reward them.
- **dismissEndModal()**
 - **What it does:** Closes the "Oops! You've reached the end" pop-up that appears when a user scrolls to the absolute rock-bottom of the website.